4DGS360: 360° Gaussian Reconstruction of Dynamic Objects from a Single Video

Jae Won Jang, Yeonjin Chang, Wonsik Shin, Juhwan Cho, Nojun Kwak | Seoul National University

https://arxiv.org/abs/2603.21618

Research Questions

- How can we reconstruct complete 360° dynamic object geometry from monocular video when occluded regions cause existing 2D-based tracking methods to fail?
- How can we design a 3D-native initialization that preserves geometry in occluded regions to enable coherent 360° 4D reconstructions?
- How do we evaluate 360° dynamic reconstruction performance when existing datasets lack sufficient view disparity between training and test cameras?

Core Idea

Existing monocular 4D reconstruction methods rely on 2D tracking to initialize 3D Gaussian trajectories, but depth maps only provide depth for visible surfaces, causing occluded regions to be poorly initialized and leading to overfitting to training views. The key insight is to use AnchorTAP3D, which leverages confident 2D track points as spatial-temporal anchors for 3D tracking, enabling reliable geometry estimation even for occluded points. This improved initialization allows optimization strategies like ARAP regularization to work properly, yielding coherent 360° reconstructions where previous methods fail at extreme viewpoints.

Key Formula

Anchor Set Formation

```
\mathcal{X}^{A}_{w(t,t')}   \;=\;   \big\{\,   \mathbf{x}^{2D}_{s'}   \;\big|\;   s'\!\in   \!   w(t,t'),\;
\text{Mask}_{s'}\!=\!1 \,\big\}
```

Defines the set of 3D anchor points X^A within temporal window w(t,t') by collecting reliable 2D-to-3D lifted points x^2D where the confidence mask equals 1. This anchor set provides geometry-consistent supervision across frames for the 3D tracker.

Anchor-Guided 3D Tracking

$$(\mathbf{x}_{t'}^{Anchor}, v_{t'}) = \mathcal{P}^{3D}_{t \to t'}(\mathbf{p}_t, \mathcal{I}, \mathcal{D}, \Theta, \mathcal{X}^{A}_{w(t,t')})$$

The core AnchorTAP3D formulation where P^3D predicts 3D position x^Anchor and visibility v at target time t' given query point p_t, images I, depths D, camera parameters Θ, and crucially the anchor set X^A. The anchor conditioning enables stable tracking through occlusions.

Key Figure

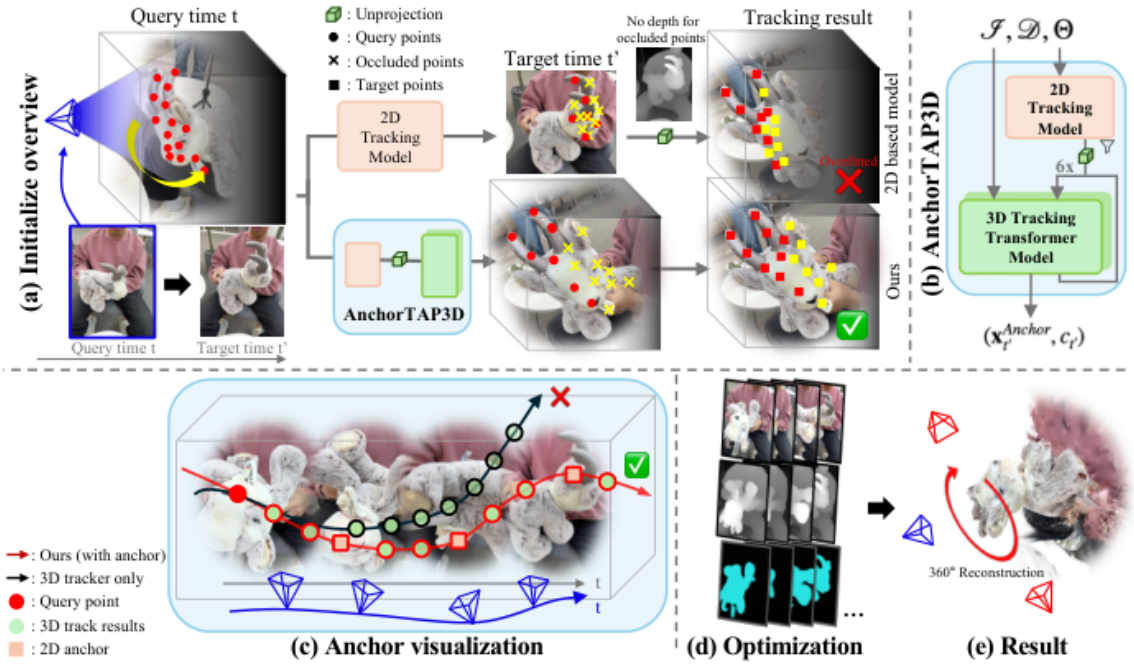

Fig. 2: Overview of 4DGS360 (a) illustrates the initialization stage using AnchorTAP3D, where reliable geometry can be obtained even when tracked points become occluded. This contrasts with 2D-based models, which cannot estimate depth for occluded points. (b) presents the overall architecture of AnchorTAP3D, and (c) highlights the difference between tracking with and without anchors, showing that anchors effectively prevent error drift. (d) depicts the training process that follows initialization, and (e) shows that our model finally enables full 360° reconstruction. The blue pyramids represent training cameras, while the red pyramids indicate test cameras.

tracking directly in XYZ space, leveraging unprojected image features to construct a spatio-temporal 3D feature cloud. It models temporal correspondence through 3D neighborhood-to-neighborhood attention, where local geometric features near query and target points are jointly attended to infer motion trajectories. This design enables TAPIP3D to capture smooth and continuous motion across frames and handle moderate occlusions without explicit depth supervision. While 3D tracking provides more geometric understanding than 2D tracking models, it is sensitive to errors in depth maps and camera calibration, and tracking failures tend to accumulate over time, reducing long-term stability. On the other hand, 2D tracking is generally more resilient to noise in real-world settings but provides limited spatial understanding. To address this, we propose **AnchorTAP3D**, which leverage high confidence 2D track points as anchors for 3D tracking, enabling consistent geometry aware initialization for our model.

3 Method

Our goal is to reconstruct a 360° 4D representation over time, given a sequence of T training frames $\mathcal{I} = \{I_t\}_{t=1}^{T}$, depth maps $\mathcal{D} = \{D_t\}_{t=1}^{T}$, and camera parameters $\Theta = \{\theta_t\}_{t=1}^{T}$ obtained either from dataset sensors or pretrained estima-

Figure 2: Shows the complete 4DGS360 pipeline including AnchorTAP3D architecture, anchor visualization demonstrating drift

prevention, and the progression from initialization through optimization to final 360° reconstruction.

Limitations

- Performance still depends on the capability of pretrained 2D and 3D tracking models used as components
- Assumes fixed color for each Gaussian over time, cannot handle illumination changes in real-world scenes
- Cannot synthesize extreme viewpoint background regions if invisible in input video
- Requires depth maps and camera parameters as input, limiting applicability to casual capture scenarios